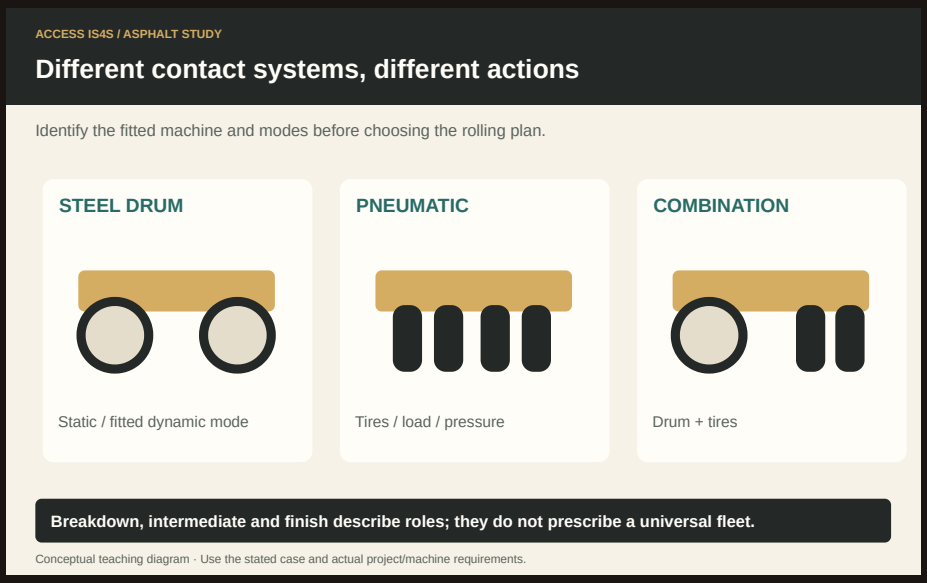
ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Compaction and final handover

Volume 8 · L1, L2, L3, L4, L5, L6

Classroom development edition · 30 October 2026



ACCESS IS4S. (2026). Different contact systems, different actions [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

In this volume

1. How to use the volume and key terms
2. Compaction equipment
3. Compaction work safety procedures
4. Asphalt Compaction Roller operator roles and responsibilities
5. Compaction Roller maintenance
6. Operate Compaction Roller
7. Follow shut down procedures
8. Read the rolling record and its limits

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for compaction and final handover. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

Study chapter · Outline L1

Compaction equipment

Compaction equipment changes the mixture's density and surface through a planned application of load and, where fitted, dynamic energy. Choose equipment to suit the mixture, support, production and finished requirements.

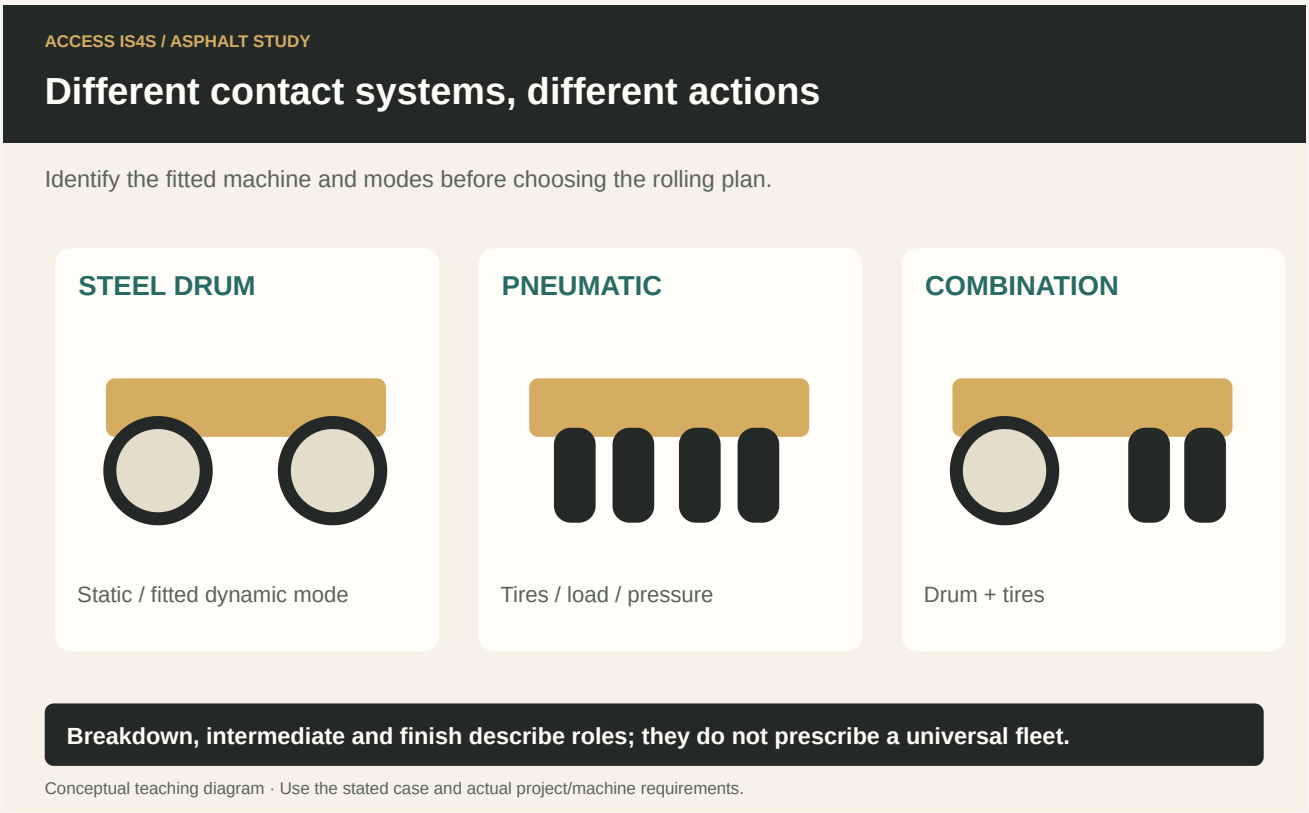
- Types of Compaction Rollers
- Major components of compaction equipment
- Basic functions of compaction equipment
- Emerging technology

LEARNING STEP 1

Types of Compaction Rollers

Manufacturers such as Caterpillar, Ammann, BOMAG and Dynapac offer different roller families and configurations. Compare drum width, operating mass, tire/drum arrangement, working modes, production and access. A small utility roller and a highway tandem roller are not interchangeable simply because both compact asphalt. Confirm the actual model, serial range and OMM. Equipment choice and the rolling plan belong to the responsible crew/quality process.

Different contact systems, different actions



Note. ACCESS IS4S. (2026). Different contact systems, different actions [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Major components of compaction equipment

A steel-drum roller may work statically or with fitted vibration/oscillation. A pneumatic-tired roller applies load through tires; tire condition, pressure, temperature and ballast affect its action under the manufacturer and project requirements. Do not invent tire pressures. A combination roller has both drum and tire contact. Identify engine, travel/steering controls, articulation joint, drums/tires, water/spray system, scrapers, vibration controls, ROPS and seat belt where fitted. Components and permitted modes vary by model.

LEARNING STEP 3

Basic functions of compaction equipment

Breakdown rolling applies early compaction behind the paver within suitable mixture conditions. Intermediate rolling develops required density and surface condition as planned. Finish rolling removes marks and completes the surface without damaging it. One machine may perform more than one role where the plan supports it; role names do not prescribe a universal machine count or pass sequence.

Locate all 19 original roller-control callouts

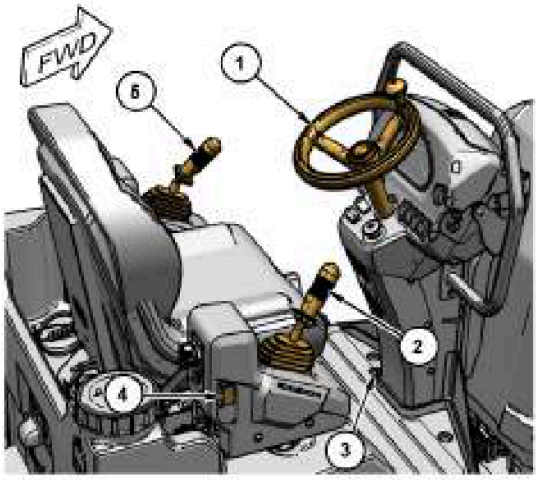


Illustration 39 g06673749

(1) Steering wheel
(2) Right propel control
(3) Traction Control Switch (if equipped)
(4) 12V power receptacle
(5) Left propel control (if equipped)

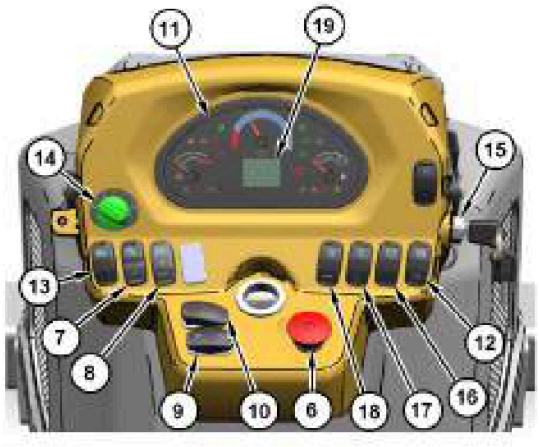


Illustration 40 g06610378

(6) Emergency stop knob
(7) Work lights
(8) Front/Position lights
(9) Turn signal
(10) Hazard lights
(11) Gauges and indicators
(12) Parking brake
(13) Horn
(14) Water spray system
(15) Engine start switch
(16) Vibratory mode / Emulsion switch
(17) Engine speed
(18) Beacon
(19) Digital display

Steering Wheel (1)

Steering wheel (1) controls the directional steering of the machine. The machine will turn in the same direction as steering wheel (1).

Note. Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022), source PDF p. 41). Faithful crop; original labels retained. CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Illustration 39 has 1–5; Illustration 40 has 6–19. Optional items retain “if equipped.” Full source context.

LEARNING STEP 4

Emerging technology

Oscillation applies a different dynamic action from conventional vertical vibration. Intelligent compaction, mapping and temperature displays can help monitor coverage and conditions. Their displayed values are not automatically the contract’s acceptance density. Confirm what each measurement represents and how it is calibrated/validated. Technology supports operator judgement and quality testing; a colourful coverage map does not prove every point meets specification.

Vibratory roller



Figure 9.7 Vibratory Roller

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 9.7, p. 430; PDF p. 464). Faithful scan crop; original figure caption retained. Full source context.

Worked example · Choose by the task

A narrow access area has restricted turning space, while the main road has a high production rate.

Think it through.

Compare suitable machine dimensions, modes, capacity and support for each area, then coordinate the planned roles and testing. Do not select only by the largest operating mass or assume one roller can maintain both areas without a scheduling conflict.

Try it and explain your thinking

Explain and apply the idea

Identify drum, scraper, spray bar, articulation joint, ROPS and seat belt on the supplied model figures.

Skills you use: Reading, Communication.

A fictional effective coverage width is 1.4 m over a 70 m strip

A fictional effective coverage width is 1.4 m over a 70 m strip. Find area for one full-length coverage strip.

Skills you use: Numeracy, Problem solving.

Ask an AI to compare static, vibratory and oscillatory action, then audit its table.

Ask an AI to compare static, vibratory and oscillatory action, then audit its table.

Skills you use: Digital, Writing, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify roller families and actual components.	0 / 1 / 2	Evidence/help: _____
Explain breakdown, intermediate and finish roles.	0 / 1 / 2	Evidence/help: _____
Distinguish monitoring displays from acceptance tests.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, L1, printed p. 93; source PDF pp. 95–95.
- Caterpillar. (Various editions). Original machine component figures, with publication/page/model records. Open publication.
- Caterpillar. (2012). Guide to Asphalt Compaction (QEBQ1583). Asphalt compaction concepts and equipment; project and machine application require the applicable documents. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 9: Hot Mix Asphalt Compaction; source PDF pp. 447–478. Open publication.
- Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022)). CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline L2

Compaction work safety procedures

A roller repeatedly changes direction near people and other equipment. Safe operation depends on visibility, ground support, communication and the machine’s protective systems throughout the pattern.

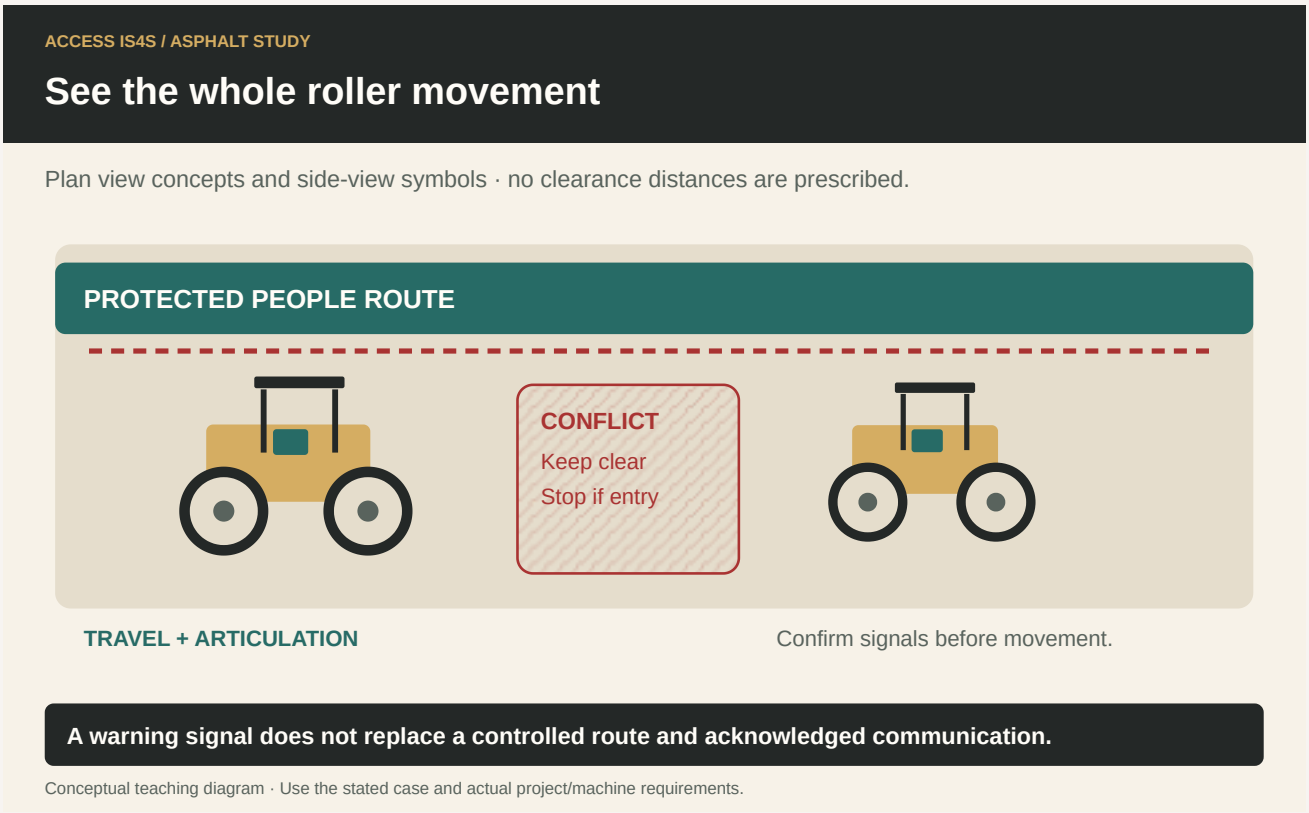
- Interpret job planning
- Machine-specific safety hazards
- Traffic control requirements
- PPE
- Lockout procedures
- Secure machine for safety

LEARNING STEP 1

Interpret job planning

Review the path, pattern, thickness, temperature conditions, crown, joints, obstacles and coordination of the asphalt train. Identify edges, slopes, structures, traffic and public access. Confirm the current plan and who authorizes a change. Walk the inspection route only under controlled conditions.

See the whole roller movement



Note. ACCESS IS4S. (2026). See the whole roller movement [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Machine-specific safety hazards

Recognize blind areas, crush/articulation zones, hot surfaces, noise, fumes, hydraulic pressure, rollover hazards and proximity to pavers/other rollers. Inspect and use the fitted ROPS and seat belt according to the OMM and applicable rules; do not modify protective structures. A fixed three-metre spacing is not a universal safe separation. Establish actual clearances from the machine instructions and site assessment, including stopping distance and visibility.

LEARNING STEP 3

Traffic control requirements

Use the traffic-control plan, trained personnel, signs and communication arrangement. Keep the public route separate from reversing/turning areas. Obtain acknowledgement before entering another machine's working zone. If a pedestrian, vehicle or unexpected obstacle enters, stop affected movement safely and restore control before continuing.

LEARNING STEP 4

PPE

Wear the selected footwear, head, eye, hearing and high-visibility protection and any task-specific items. Ensure the belt and operator position remain compatible with safe control. PPE does not replace the protective structure, separation or visibility checks. Supply, use and maintenance responsibilities follow current Part 8 and employer arrangements.

LEARNING STEP 5

Lockout procedures

Service and hazardous access require the actual machine's lockout, stored-energy and support procedure. The articulation joint can move and crush; use the prescribed steering-frame lock where required and only in the specified conditions. A parked engine does not remove every hydraulic or gravity hazard. Verify the safe state before work.

LEARNING STEP 6

Secure machine for safety

Park on the approved stable ground, apply the specified parking controls and required chocks, and secure keys/access. Position away from unsupported edges and public routes. Record defects and protect people from a machine that could roll or be moved by an unauthorized person. A battery/night switch is only one part of the actual security/isolation arrangement.

Articulation and the steering frame lock

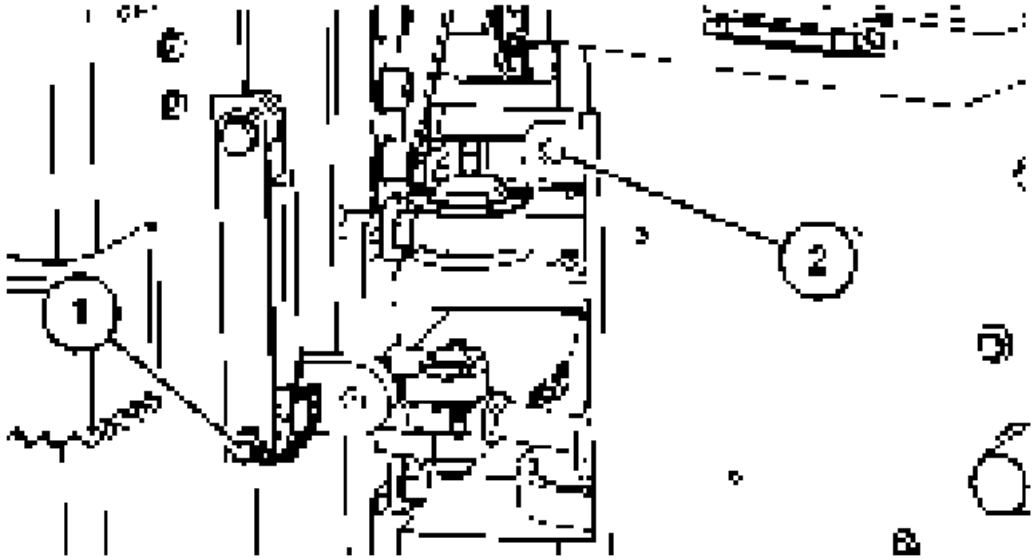


Illustration 34 g06183886

Unlocked position

Note. Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022), source PDF p. 38). Faithful crop; original labels retained. CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Read the complete source page for locking/unlocking instructions; this task is identification only. Full source context.

Worked example · A shortcut behind the roller

A worker steps behind a roller to shorten the route while the operator is preparing to reverse.

Think it through.

Stop the affected movement and re-establish the controlled pedestrian route and communication. Review the blind-area conflict with the crew. High-visibility clothing does not authorize crossing an active reversing path.

Try it and explain your thinking

Explain why “three metres is always safe” is an inadequate rule.

Explain why “three metres is always safe” is an inadequate rule.

Skills you use: Problem solving, Reading.

A route change reduces available clearance by 0.6 m

A route change reduces available clearance by 0.6 m. What should be recorded?

Skills you use: Numeracy, Writing.

Demonstrate a clear stop and read-back with the paver lead.

Demonstrate a clear stop and read-back with the paver lead.

Skills you use: Communication, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Inspect/use fitted ROPS and belt and recognize blind/articulation zones.	0 / 1 / 2	Evidence/help: _____
Demonstrate the applicable lockout and frame-lock procedure.	0 / 1 / 2	Evidence/help: _____
Park and secure with required chocks/controls and access protection.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, L2, printed p. 94; source PDF pp. 96–97.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 8: Personal Protective Clothing and Equipment. Open publication.
- Caterpillar. (2012). Guide to Asphalt Compaction (QEBQ1583). Asphalt compaction concepts and equipment; project and machine application require the applicable documents. Open publication.
- Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022)). CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline L3

Asphalt Compaction Roller operator roles and responsibilities

The roller operator coordinates density development and surface finish with the paving train. A useful rolling pattern is planned, checked and adjusted through evidence rather than repeated by habit.

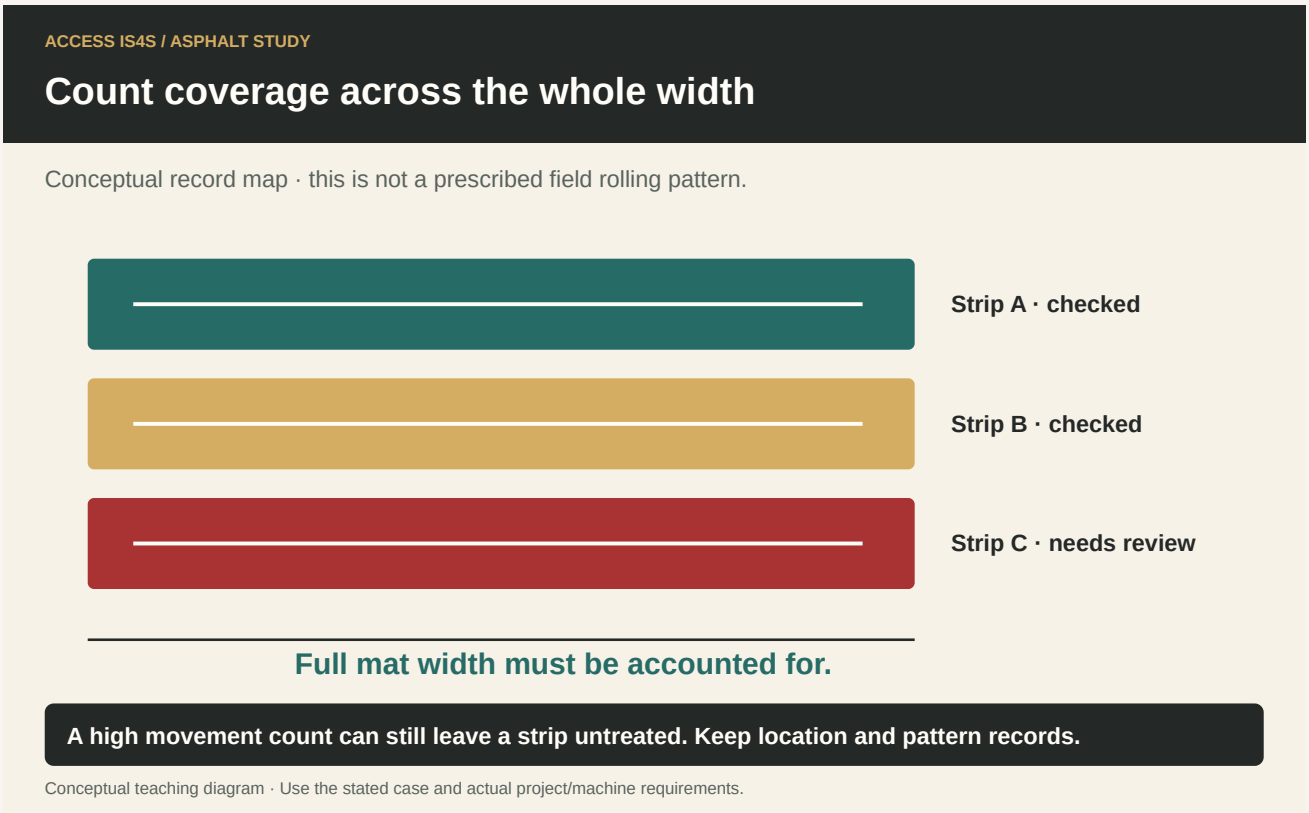
- Communication techniques
- Effective rolling pattern
- Determine pattern to achieve ride planning
- Responsibilities of Compaction Roller operator

LEARNING STEP 1

Communication techniques

Agree machine roles, lane/area, direction-change locations, signal/radio channel and stop response. With multiple rollers, share which area is complete, which still needs work and where testing or handwork is occurring. Use location and pass/coverage terminology consistently. An unacknowledged instruction is not a completed handover.

Count coverage across the whole width



Note. ACCESS IS4S. (2026). Count coverage across the whole width [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Effective rolling pattern

A rolling pattern specifies sequence, overlap, operating mode, speed and repetitions for the actual mixture and conditions. A pass and a coverage must be defined: one machine movement is not necessarily one complete treatment of the full mat width. Establish the pattern with the approved trial/quality process and record changes. Repeatedly rolling one strip while missing another creates uneven treatment even if the total pass count looks high.

LEARNING STEP 3

Determine pattern to achieve ride planning

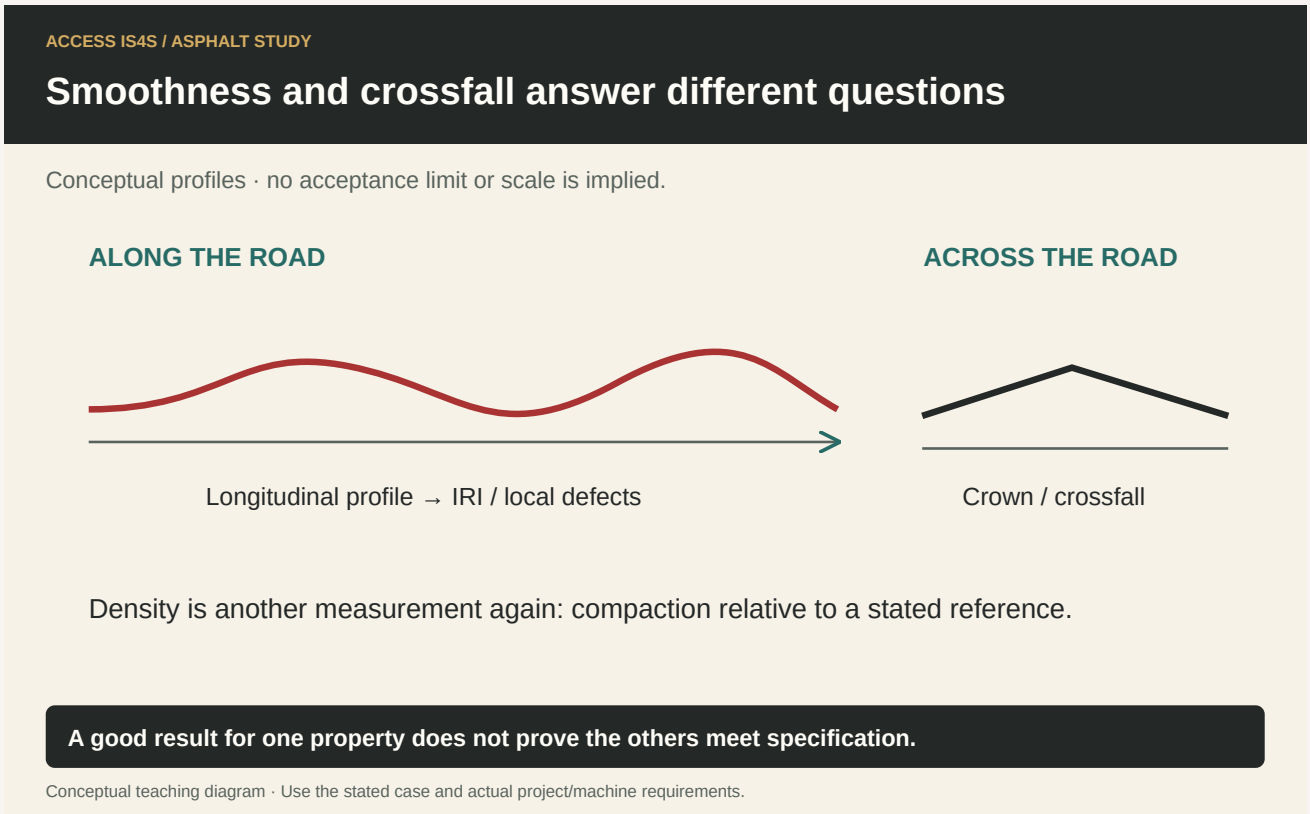
Plan for ride quality before the mat cools. Coordinate steady paving, joints, starts/stops and smooth direction changes. Abrupt turns or stopping in a damaging condition can mark or displace the mat. The finish roller cannot be assumed to repair every earlier profile defect. Use measured feedback and report persistent marks or instability promptly.

LEARNING STEP 4

Responsibilities of Compaction Roller operator

The operator monitors surroundings, mixture condition, spray/scraper performance, working mode and the planned pattern. Avoid sharp cuts across the mat and do not allow vibration to continue while stationary where prohibited by the machine/process instructions. Coordinate changes with the lead instead of adding passes independently. Required density, smoothness and surface condition are separate checks; meeting one does not establish the others.

Smoothness and crossfall answer different questions



Note. ACCESS IS4S. (2026). Smoothness and crossfall answer different questions [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Pass count without coverage

Two operators each report six passes, but their paths overlap on one side and leave a strip untreated.

Think it through.

Reconstruct the actual path and coverage map, identify the missed strip and obtain an authorized response while conditions allow. Define pass/coverage consistently for the next run. Twelve reported movements do not establish uniform compaction.

Try it and explain your thinking

Draw a conceptual coverage map with three labelled strips and a legend for complete/needs check.

Draw a conceptual coverage map with three labelled strips and a legend for complete/needs check.

Skills you use: Writing, Creativity and innovation.

A training record has four strips, each requiring two planned coverages

A training record has four strips, each requiring two planned coverages. How many strip-coverages are planned?

Skills you use: Numeracy, Problem solving.

Write a handover for a strip awaiting a quality check.

Write a handover for a strip awaiting a quality check.

Skills you use: Communication, Collaboration, Digital.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Coordinate multiple operators using clear location and coverage terms.	0 / 1 / 2	Evidence/help: _____
Follow and record the approved pattern.	0 / 1 / 2	Evidence/help: _____
Recognize and report ride/texture concerns without unsupported corrections.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, L3, printed p. 96; source PDF pp. 98–98.
- Caterpillar. (2012). Guide to Asphalt Compaction (QEBQ1583). Asphalt compaction concepts and equipment; project and machine application require the applicable documents. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 3: Rights and Responsibilities. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 9: Hot Mix Asphalt Compaction; source PDF pp. 447–478. Open publication.
- Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022)). CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline L4

Compaction Roller maintenance

Roller maintenance supports safe control and consistent contact with the mat. Water sprays, scrapers and clean contact surfaces are process equipment as well as housekeeping items.

- Service lubrication systems (greasing)
- Service air intake system as required (air filters/pre cleaners)
- Drain fuel tank sump, water separator fuel filters is equipped
- Add fluids as required
- Inspect and clean components as required
- Housekeeping tasks

LEARNING STEP 1

Service lubrication systems (greasing)

Use the actual lubrication chart and fitted system. Confirm product, interval, quantity and safe position, clean fittings and load the grease gun correctly. Articulation-area access requires the specified protection. Do not transfer a grease quantity from a different roller or force a blocked fitting.

Keep the contact surface working consistently



Note. ACCESS IS4S. (2026). Keep the contact surface working consistently [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Service air intake system as required (air filters/pre cleaners)

Read intake restriction indicators and service the pre-cleaner/filter through the manufacturer’s method. Inspect seals and prevent contamination of the clean side. Clean only when permitted; otherwise replace the element. An employer’s stricter replacement policy may apply. Do not assume reduced-pressure air is acceptable for every element.

LEARNING STEP 3

Drain fuel tank sump, water separator fuel filters is equipped

Where fitted, drain the fuel sump/water separator under the prescribed conditions, collect the liquid and report water/debris. Control ignition and exposure. Keep this routine task separate from high-pressure fuel-system repairs. Record unusual or repeated contamination.

LEARNING STEP 4

Add fluids as required

Identify the correct fluid and reservoir, then use the specified level ground, temperature and engine state for checking. Dual-scale dipsticks require the correct reading condition. Clean fill tools and avoid mixing, overfill and dirt/water entry. Report repeated losses and leaks rather than treating top-up as a permanent correction.

LEARNING STEP 5

Inspect and clean components as required

Inspect radiator/oil cooler, plugs, dipsticks, batteries, hoses, guards and accessible components. Protect fins and electrical parts during cleaning. Hot coolant, pressurized hydraulics and battery hazards require their actual procedures. Inspect safety systems and report defects for authorized repair. A maintenance checklist does not authorize work beyond the learner's training.

LEARNING STEP 6

Housekeeping tasks

Remove debris and excess grease from safe access areas. Maintain the spray tank, filters, nozzles and approved water-treatment arrangements, and inspect drum/tire cleanliness, scrapers and fitted mats. Blocked sprays can allow pickup and mark the surface. Prepare the water system for freezing conditions using the model's drain/winterization procedure; "drain all water" requires knowing every affected component. Collect waste, restore guards and record service and defects.